

Decoding101: A beginner's guide to decoding

<https://www.youtube.com/@TenMinuteKQL>

Table Name	Description	Example
 AuthenticationEvents	Logins to devices (successes and failures).	When someone tries to log in to their email account.
 Email	Emails sent and received by employees.	When an employee sends or receives a work email.
 Employees	Info about the company's employees.	Names, job titles, and contact details of employees.
 FileCreationEvents	Files created on employee devices.	When someone saves a new document on their computer.
 InboundNetworkEvents	Internet activity coming into the company network.	When someone visits the company website.
 OutboundNetworkEvents	Internet activity going out from the company network.	When someone visits an external website like google.com.
 PassiveDns (External)	Which IP addresses match up with which websites.	Figuring out that "google.com" matches with IP "8.8.8.8".
 ProcessEvents	Programs that start running on employee devices.	When an employee opens a program like Microsoft Word.
 SecurityAlerts	Security warnings from devices or email systems.	When antivirus software detects a virus on a computer.

Table very useful to be translate to other sequel language

KQL examples

```
Employees
| count

Email
| where recipient == "noemi_tep@jojoshospital.org"
| where sender !contains "jojoshospital"

AuthenticationEvents
| where src_ip == "10.10.0.144" and result == "Failed Login"

How many results do we get in AuthenticationEvents for these two IP addresses?
AuthenticationEvents
| where src_ip == "10.10.0.144" or src_ip == "10.10.0.86"
```

How many failed login attempts were from these IP addresses?

```
AuthenticationEvents
| where src_ip == "10.10.0.144"
or src_ip == "10.10.0.86"
or src_ip == "10.10.0.86"
or src_ip == "10.10.0.20"
or src_ip == "10.10.0.109"
```

```
AuthenticationEvents
| where src_ip in ("10.10.0.144", "10.10.0.86", "10.10.0.86", "10.10.0.20",
"10.10.0.109")
```

```
AuthenticationEvents
| where src_ip in ("10.10.0.144", "10.10.0.86", "10.10.0.86", "10.10.0.20",
"10.10.0.109") and result == "Failed Login"
```

How many distinct websites(urls) did Nancy Roberts visit?

```
Employees
| where name == "Nancy Roberts"
```

```
OutboundNetworkEvents
| where src_ip == "10.10.0.30"
| distinct url
```

```
// Finding Nancy Roberts' IP address to check web browsing
Employees
| where name == "Nancy Roberts"
//Nancy's ip address: 10.10.0.30
```

```
// Find all IT support employees
Employees
| where role contains "IT support"
```

```
// Look for logins from IPs used by IT support
AuthenticationEvents
| where src_ip in ("10.10.0.75", "10.10.0.42", "10.10.0.34", "10.10.0.10", "10.10.0.2")
```

```
// Count all failed logins
AuthenticationEvents
| where result == "Failed Login"
| count
```

How many websites did employees with the first name Mary visit?

Methode 1

```
// search employee with first name mary
Employees
| where name contains "Mary"
```

```
OutboundNetworkEvents
```

```
| where src_ip in ("10.10.0.84", "10.10.0.169", "10.10.0.161", "10.10.0.24",  
"10.10.0.125", "10.10.0.113", "10.10.0.103")  
| count
```

Methode 2

```
let mary_ips = Employees  
| where name has "Mary"  
| distinct ip_addr;  
OutboundNetworkEvents  
| where src_ip in (mary_ips)
```

How many authentication attempts did we see to the accounts of employees with the first name Mary?

```
let Mary_name = Employees  
| where name has "Mary"  
| distinct username;  
AuthenticationEvents  
| where username in (Mary_name)  
| count
```